TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Sonia Chernova	**Email**	chernova@gatech.edu
Institute Name	Georgia Institute of Technology			
Address	Atlanta, Georgia, USA			

2. Technology Overview

Title of Proposed Technology	**Vision–Language–Grounded Task and Motion Planning for Robust, Human-Centered Robot Autonomy**				
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), **AI (√),** Next-gen Communication ( ), **Advanced Robotics & Manufacturing (√),** Quantum ( )				
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)			**Technology Field 2** (Large Category)	
	Mechanical · Materials			Mechanical · Materials	
	Medium Category		Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies	Robotics/ Automation Machinery	Robot Vision and Production Automation Technologies	Robotics/ Automation Machinery
Concept of Proposed Technology	The proposed technology aims to develop an integrated robotic planning and reasoning framework that enables robots to operate autonomously over long time horizons while interacting fluently and transparently with humans. The system unifies **long-horizon Task and Motion Planning (TAMP)** grounded in **Vision–Language Models (VLMs)** with **proactive robot behavior** and **explainable decision-making**, allowing robots to plan, anticipate, and explain multi-step tasks in dynamic, real-world environments. At the core of the proposed technology is a VLM-grounded TAMP framework that leverages semantic understanding from vision and language to guide planning under uncertainty. VLMs extract high-level				

	contextual information about objects, spaces, and tasks, which is integrated into symbolic task representations and continuous motion planning. This enables the robot to reason jointly about what to do and how to do it, supporting robust execution of extended task sequences that require perception, navigation, and manipulation across changing environments. To reduce user burden and improve interaction fluency, the proposed technology incorporates proactive robot behavior directly into the planning process. Rather than relying solely on explicitly specified goals, the robot learns predictive models of user intent, preferences, and routine patterns from interaction history and environmental observations. These models allow the planner to anticipate future needs and incorporate supportive actions into its plans, enabling the robot to assist users without requiring constant instruction while preserving human oversight and control. As robots plan over longer horizons and take greater initiative, explainability becomes essential for maintaining trust and situational awareness. The proposed technology embeds explanation mechanisms within the planning architecture itself, enabling the robot to generate faithful, human-interpretable explanations of its actions. By exposing planning alternatives, uncertainty, and anticipated outcomes, the system can communicate why specific actions were chosen, how proactive decisions relate to inferred user goals, and how uncertainty influences behavior. Together, these capabilities will form a unified technology that advances robot autonomy beyond short-horizon, reactive execution. By integrating VLM-based semantic reasoning, long-horizon TAMP, proactive assistance, and explainable AI within a single framework, the proposed technology enables robust, usable, and trustworthy robotic systems capable of supporting humans across extended tasks in complex environments.
Rationale for Joint R&D Project	The proposed research requires close collaboration between academic research labs at Georgia Tech and an industrial partner to translate advances in long-horizon Task and Motion Planning, proactive robot behavior, and explainable AI into deployable robotic systems. While core algorithms can be developed in simulation and laboratory settings, their robustness, safety, and usability must be validated through integration with real robotic platforms operating under realistic constraints. Georgia Tech's RAIL research lab, directed by Dr. Chernova, will provide expertise in VLM-grounded planning, proactive assistance, and explainable sequential decision-making, while the industry partner will contribute robot hardware development, integration, and real-world experimentation capabilities. A joint R&D framework enables iterative hardware–software co-development, reducing technical risk and accelerating the transition from research prototypes to practical systems. This collaboration is essential to ensure that the proposed technologies support robust long-horizon execution, effective human–robot interaction, and scalable deployment in real-world environments.
Key Words	1. Vision–Language–Grounded Task and Motion Planning 2. Proactive Robot Assistance 3. Explainable Decision-Making for Robotics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The objective of this R&D project is to develop a robust and usable robotic planning framework that enables robots to execute **long-horizon tasks in dynamic, human-centered environments** through the integration of **Vision–Language Models (VLMs)**, **Task and Motion Planning (TAMP)**, **proactive robot behavior**, and **explainable decision-making**. The specific R&D objectives are as follows: 1. **Develop VLM-Grounded Long-Horizon TAMP:** Design and implement a task and motion planning framework that integrates semantic reasoning from vision and language with symbolic and continuous planning to support multi-step task execution under uncertainty. 2. **Enable Robust Long-Horizon Task Execution:** Develop adaptive planning, execution monitoring, and recovery mechanisms that allow robots to maintain task progress over extended horizons despite environmental changes, partial observability, and execution failures. 3. **Integrate Proactive Robot Behavior:** Develop predictive models of user intent and preferences and incorporate them directly into planning objectives, enabling anticipatory actions that reduce user burden while preserving human oversight. 4. **Support Explainable Robot Decision-Making:** Embed explainable AI mechanisms within the planning architecture to generate faithful, human-interpretable explanations of robot actions, including the role of uncertainty, alternatives, and proactive decisions. Together, these objectives aim to advance the state of the art in human-centered robot autonomy by enabling robots that can plan, anticipate, and explain their behavior over long time horizons in real-world environments.			
R&D Target Specifications	The success of this R&D project will be evaluated based on the following criteria: **Long-Horizon Planning and Execution:** Ability to plan and execute multi-step tasks robustly in dynamic, partially observable environments, including adaptive replanning during execution. **VLM-Grounded Perception and Reasoning:** Effective grounding of language and visual information into task and motion planning representations for reliable object- and context-aware decision-making. **Proactive Robot Behavior:** Demonstrated ability to anticipate user needs and incorporate supportive, context-appropriate actions that reduce user burden. **Explainability and Usability:** Ability to provide faithful, understandable explanations of robot behavior and support intuitive interaction for non-expert users.			

R&D Plan	The R&D project will be conducted in three phases, progressing from algorithm development to system integration and validation in realistic environments. **Phase 1: Core Algorithm Development (Year 1)** ● Develop a VLM-grounded long-horizon Task and Motion Planning framework that integrates semantic reasoning with symbolic and continuous planning. ● Implement execution monitoring and adaptive replanning mechanisms to support robust long-horizon task execution. ● Develop initial models for proactive robot behavior based on user activity and environmental context. ● Milestone: Demonstration of long-horizon planning and proactive behavior in simulation-based task scenarios. **Phase 2: System Integration and Laboratory Validation (Year 2)** ● Integrate planning, perception, proactivity, and explainability modules on a mobile manipulation platform. ● Develop natural-language interfaces for task specification and explanation of robot behavior. ● Conduct laboratory-scale experiments to evaluate robustness, proactivity, and explainability under controlled conditions. ● Milestone: Integrated robotic system demonstrating robust, proactive, and explainable task execution in laboratory environments. **Phase 3: Real-World Evaluation and Refinement (Year 3)** ● Deploy the integrated system in realistic, dynamic environments representative of service robotics scenarios. ● Evaluate long-horizon execution, proactive assistance, and explanation quality through extended task trials and user studies. ● Refine system performance, usability, and reliability based on experimental results. ● Milestone: Validated prototype demonstrating long-horizon, proactive, and explainable robotic task execution suitable for real-world deployment.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Georgia Tech RAIL Lab will be responsible for the development of the core AI methodologies and system-level intelligence underpinning the proposed technology. This includes leading the research on long-horizon Task and Motion Planning, Vision–Language–Model-based reasoning, proactive robot behavior, and explainable decision-making. Specific roles and responsibilities include: ● Design and development of a VLM-grounded long-horizon TAMP framework, including adaptive planning and execution monitoring mechanisms. ● Development of proactive robot behavior models, including user-intent and context prediction, and their integration into planning objectives. ● Design and implementation of explainable planning and decision-making mechanisms that generate faithful, human-interpretable explanations of robot behavior. ● Development of simulation environments, benchmarking protocols, and experimental methodologies for evaluating robustness, proactivity, and explainability. ● Technical leadership and close collaboration with the industrial partner to support system integration, experimental validation, and iterative refinement. Through these responsibilities, the research institute will ensure the scientific rigor, technical soundness, and human-centered design of the proposed robotic planning system.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for: ● Design of the hardware with specific sensing capabilities or provide the hardware if this already exists. ● Integration of developed software for testing at scale. ● Testing and validation of the pipeline in various scenarios ● Feedback and guidance on revision and refinement of the system

5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Development of a validated robotic planning framework that integrates Vision–Language Models, long-horizon Task and Motion Planning, proactive behavior, and explainable decision-making. 2. Demonstration of a prototype robotic system capable of robust multi-step task execution and anticipatory assistance in dynamic, real-world environments. 3. Advances in human-centered robot autonomy, including methods for transparent and interpretable explanation of long-horizon and proactive robot behavior. 4. Transferable technologies and system designs that support service robots in both household and business environments. 5. Strengthened research–industry collaboration and accelerated pathways toward commercialization of AI-driven service robotics solutions.